## Switchgear

Switchgear is a protective device used for switching "ON" and "OFF" of generators, transmission lines, distribution lines and other equipment under both normal and abnormal conditions. A switchgear is a centralized collection of switching and protective devices such as switches, fuses, circuit breakers, relays etc. The collection of these separate devices are mounted in a common metal enclosure. During normal operation, switchgear permits to switch "ON" generators, transmission lines, distribution lines and other electrical equipment. On the other hand, when a fault (eg. short circuit) occurs on any part of power system, heavy current flows through the equipment, threatening damage to the equipment and interruption of service to the customers. However, the switchgear detects the fault and disconnects the unhealthy section from the power network. In this way, switchgear protects the power network from damage and ensures continuity of supply.

## Switchgear Selection

Switchgear is selected after consideration of the following;

(1) Select a switchgear whose voltage rating is at least 20% higher than the voltage of the circuit where the switchgear is to be installed.

(2) The switchgear current rating in Kilo-Ampere (KA) must be at least 10% higher than the continuous current of the circuit where the switchgear is to be installed.

(3) The short-circuit current rating of the switchgear selected must be slightly higher than the fault current ($I_f$) of the circuit.

(4 Service conditions of the switchgear like altitude, ambient temperature, humidity must be considered.

## Circuit Breaker

Circuit breaker is an automatically operated switch designed to protect an electrical circuit from damage caused by excess current from an overload or short circuit.

## Circuit Breaker Selection

(1) The voltage of the circuit breaker must exceed the maximum voltage at which the system will operate.

(2) The continuous current rating of the circuit breaker must ~~exceed~~ maximum continuous current of the circuit where the circuit breaker will operate.

(3) Circuit breaker short circuit current, must equal or slightly exceed available fault current ($I_f$).

## Busbars

Busbars are copper or aluminium rods or thin walled aluminium or copper tubes with approximately zero resistance and operate at constant voltages. Busbars are used to connect together a number of generators, transmission / distribution lines, feeders or transformers operating at the same voltage.

## Busbar Selection

In busbar selection, the first step shall be to calculate the minimum cross-section area of single busbar to carry the fault current. The following formula shall enable us to calculate the minimum cross-section-area (CSA) of busbar to accomodate a given fault current ($I_F$);

$$\text{Cross sectional area } (A) = \left(\frac{I_F}{0.123}\right) \times \sqrt{t} \ (mm^2) \text{ for copper busbar} \quad \ldots \ldots (20)$$

$$\text{Cross-section-area } (A) = \left(\frac{I_F}{0.07}\right) \times \sqrt{t} \ (mm^2) \text{ for aluminium busbar} \quad \ldots \ldots (21)$$

where $A$ = cross-section-area of busbar,
$t$ = Fault duration in seconds,
$I_F$ = short circuit or fault current.

Thus, busbar selection is influenced by fault duration. For complete discrimination $t = 1$ second. Other fault duration of 2-seconds, 5-seconds, 10 to 30-seconds are also available.

**Example:**

Given a fault level of 35KA and fault duration of 1second, calculate the cross-section-area of the busbars for aluminium and copper respectively.

**Solution:**

Given: fault level ($I_F$) = 35KA
fault duration ($t$) = 1second, then,

For aluminium busbar, $A = \left(\frac{35}{0.07}\right) \times \sqrt{1} = \frac{35}{0.07} = 500mm^2$

Hence, aluminium busbar of 100mm x 5mm is suitable.

For copper busbar, $A = \left(\frac{35}{0.123}\right) \times \sqrt{1} = \frac{35}{0.123} = 285mm^2$

Hence, copper busbar of 57mm x 5mm is suitable.

## Fuses

Fuse is an electrical safety device used for protecting electrical equipment and personnel against overcurrent. A fuse is made of a thin wire of low melting point housed in a glass, porcelain or plastic enclosure. If a fault occurs in the system and an over current or excess current flows through the circuit, the fuse automatically melts and breaks the contact of the circuit. Thus, protecting the appliance from any damage.

## Fuse Selection

(1) The voltage rating of the fuse must exceed the maximum voltage of the circuit where the fuse is to operate.

(2) The continuous current rating of the fuse must exceed slightly the continuous current rating of the circuit where the fuse will operate.

(3) Fuse is selected by rating it at 125% (or 1.25) of the normal operating current of the circuit. The table below illustrates how fuse rating is selected.

| S/N | Normal Operating Current Amperes (A) | Fuse Rating in Amperes (A) Normal operating current × 1.25 |
|---|---|---|
| 1 | 12 | 15 |
| 2 | 16 | 20 |
| 3 | 24 | 30 |
| 4 | 48 | 60 |
| 5 | 80 | 100 |
| 6 | 160 | 200 |
| 7 | 320 | 400 |
| 8 | 480 | 600 |
| 9 | 640 | 800 |
| 10 | 960 | 1200 |
| 11 | 1280 | 1600 |

## 3.0 Corona

When an alternating potential difference is applied across two conductors whose spacing is large as compared to their diameters, there is no apparent change in the condition of atmospheric air surrounding the wires if the applied voltage is low. However, when the applied voltage exceeds a certain value, called "critical disruptive voltage", the conductors are surrounded by a faint violet glow called corona. The phenomenon of violet glow, hissing noise and production of ozone gas in an overhead transmission line is known as "corona".

## 3.1 Factors Affecting Corona

The following are the factors upon which corona depends;

1. Atmospher: as corona ~~is~~ formed due to ionisation of air surrounding the conductors, therefore, it is affected by the physical state of atmosphere. In stormy weather, the number of ions is more than ~~in~~ normal and ~~as~~ as such corona occurs at much less voltage as compared with fair weather.
2. Conductor Surface:. the corona effect depends upon the shape and conditions of the conductors. The rough and irregular surface will give rise to more corona because unevenness of the surface decreases the value of breakdown voltage. Thus a stranded conductor has irregular surface and hence gives rise to more corona than a solid conductor.
3. Spacing between conductors: If the spacing between the conductors is made very large as compared to their diameters, there may not be any corona effect. It is because larger distance between conductors reduces electrostatic stresses at the conductor surface, thus avoiding corona
4. Line Voltage: The line voltage greatly affects corona. If it is low, there is no change in the condition of air surrounding the conductors and hence no corona is formed.

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However, if the line voltage has such value that electrostatic stresses developed at the conductor surface make the air around the conductor conducting, then corona is formed.

## 3.2 Effects of Corona

The effects of corona are;

1. Due to corona formation, the air surrounding the conductor becomes conducting and hence virtual diameter of the conductor is increased. The increased diameter reduces the electrostatic stresses between conductors.
2. Corona reduces the effects of transients produced by surges.
3. Corona is accompanied by a loss of energy. This affects the transmission efficiency of the line.
4. Ozone is produced by corona and may cause corrosion of the conductor due to chemical action.
5. The current drawn by the line due to corona is non-sinusoidal and hence non-sinusoidal voltage drop occurs in the line. This may cause inductive interference with neighbouring communication lines.

## 3.3 Critical Disruptive Voltage

This is the minimum phase-neutral voltage at which corona occurs.

Consider two conductors of radii $r$ (cm) and spaced $d$ (cm) apart. If $V$ is the phase-neutral potential, then potential gradient at the conductor surface is given by:

$$g = \frac{V}{r \log_e d/r} \text{ volts/cm}$$

In order that corona is formed, the value of $g$ must be equal to the breakdown strength of air. The breakdown strength of air at 76cm pressure and temperature of 25°C is 30kV/cm (max) or 21.2 KV/cm (r.m.s) and is denoted by $g_o$. If $V_c$ is the phase-neutral potential required under these conditions, then,

$$g_o = \frac{V_c}{r \log_e d/r}$$

where $g_o$ = breakdown strength of air at 76 cm of mercury
= 30 KV/cm (max) or 21.2 KV/cm (r.m.s)

$\therefore$ Critical disruptive voltage, $V_c = g_o r \log_e d/r$

The above-expression for disruptive voltage is under standard conditions i.e, at 76 cm of Hg and 25°C. However, if these conditions vary, the air density also changes, thus altering the value of $g_o$. The value of $g_o$ is directly proportional to air density. Thus the breakdown strength of air at a barometric pressure of b (cm) of mercury and temperature of t°C is $\delta g_o$ where

$$\delta = \text{air density factor} = \frac{3.926}{273+t}$$

Under standard conditions, the value of $\delta = 1$

$\therefore$ Critical disruptive voltage, $V_c = g_o \delta r \log_e d/r$

Correction must also be made for the surface condition of the conductor. This is accounted for by multiplying the above expression by irregularity factor $m_o$.

$\therefore$ Critical disruptive voltage, $V_c = m_o g_o \delta r \log_e d/r$ KV/phase.

where

$m_o$ = 1 for polished conductors
= 0.98 to 0.92 for dirty conductor
= 0.87 to 0.8 for stranded conductor

Example

A 3-phase line has conductors 2 cm in diameter spaced equally 1 m apart. If the dielectric strength of air is 30 KV (max) per cm, find the disruptive critical voltage for the line. Take air density factor $\delta = 0.952$ and irregularity factor $m_o = 0.9$.

## Solution

Conductor radius, $r = 2/2 = 1\,cm$

conductor spacing, $d = 1m = 100\,cm$

Dielectric strength of air, $g_o = 30\,kV/cm\,(max) = 21.2\,kV\,(r.m.s)$ per cm.

Disruptive critical voltage, $V_c = m_o g_o \delta r \log_e (d/r)$ kV/phase (r.m.s)

$$= 0.9 \times 21.2 \times 0.952 \times 1 \times \log_e (100/1)$$

$$= 83.64\ kV/phase.$$

$\therefore$ Line voltage (r.m.s) $= \sqrt{3} \times 83.64 = \underline{144.8\,kV}$

3.4 ## Visual Critical Voltage

It is the minimum phase-neutral voltage at which corona glow appears all along the line conductors. The phase-neutral effective value of visual critical voltage is given by the following expression:

Visual critical voltage, $V_v = m_v g_o \delta r \left[1 + \frac{0.3}{\sqrt{\delta \cdot r}}\right] \log_e \left(\frac{d}{r}\right)$ kV/phase.

where $m_v$ is another irregularity factor having a value of 1.0 for polished conductors and 0.72 to 0.82 for rough conductors.

3.5 ## Causes of Over-voltages in Power Systems

The causes of overvoltages on a power system may be broadly divided into two main categories viz;

1. Internal causes
2. External causes.

3.5.1 ### Internal Causes

Internal causes of overvoltages on the power system are primarily due to oscillations set up by the sudden changes in the circuit conditions. This circuit change may be a normal switching operation such as opening of a circuit breaker, or it may be the fault condition such as grounding of a line conductor.

1. Switching Surges : The overvoltages produced on the power system due to switching operations are known as switching surges.

2. Insulation Failure : the most common case of insulation failure in a power system is the grounding of conductor (i.e. insulation failure between line and earth) which may cause overvoltages in the system. The earthing of the line causes two equal voltages of $-E$ and $E$ which travel on opposite directions ~~of~~ the line from the point of the fault. These voltages produce currents which pass through the fault point to earth with magnitude equal to $2E/Z_n$ where $Z_n$ is the natural impedance of the line.

3. Arcing Ground

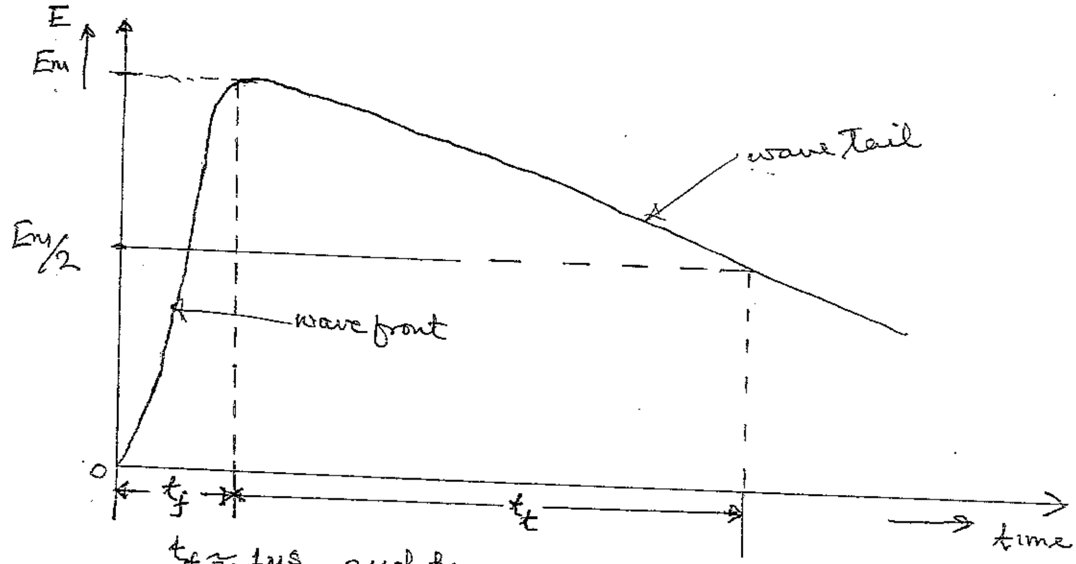
The phenomenon of intermittent arc taking place in line-to-ground fault of a 3-phase with consequent production of transients is known as arcing ground. The arcing ground produces severe oscillations of three to four times the normal voltage. The transients produced due to arcing ground are commulative and may cause serious damage to the equipment in the power system by causing breakdown of insulation.

4. Resonance : resonance in an electrical system occurs when inductive reactance of the circuit becomes equal to capacitive reactance. Under resonance, the impedance of the circuit is equal to the resistance of the circuit and the power factor is unity. This condition of resonance causes high voltages in electrical power system

5.2 External Causes

The only external cause of overvoltages on power system is lightning. An electric discharge between cloud and earth, between clouds or between the charge centres of the same cloud is known as "lightning". Lightning discharge may have currents in the range of 10KA to 90KA. A direct or indirect lightning stroke on a transmission line produces a steep-fronted voltage wave on the line. The voltage of this wave may rise from zero to peak value (perhaps 2000KV) in about 1μS and decay to half the peak value in about 5 to 50μS. Such a steep-fronted voltage wave will initiate travelling waves along the line in both directions with velocity dependent upon the L and C parameters of the line.

## 3.6 Surge Waveform




$t_f \simeq 1\mu s$ and $t_t \simeq 5\mu s$ to $50\mu S$.

Fig. 3.6 Typical surge waveform.

Fig. 3.6 shows a typical surge waveform. Voltage surge is a sudden rise in voltage for a very short duration on the power system. It is also called transient voltage. This waveform is described by its front time which is about, $t_f = 1\mu s$ and the tail time ($t_t$) which about $5\mu s$ to $50\mu S$.

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89

94.36

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## 1.10.10 Calculation of Load Flow Analysis of Interconnected System

Given a three-bus network as shown in Fig.1.10.10, the nodal admittance matrix is as shown in equation (1). The expressions for calculating power (S) at each of the buses are as given in equations (2) to (4).

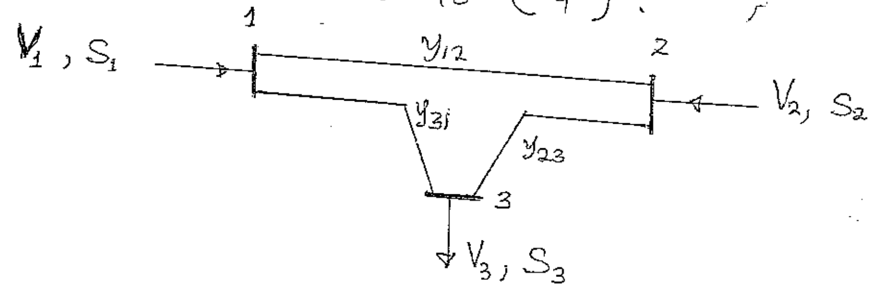



Fig.1.10.10: 3-bus Interconnected System

The bus-admittance matrix of Fig.1.10.10 is as given below;

$$Y_{bus} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & Y_{23} \\ Y_{31} & Y_{32} & Y_{33} \end{bmatrix} \quad \text{..........} (1)$$

The expression for calculating power (S) at ~~the buses~~ are;

$$S_1 = V_1\left[Y_{11}V_1 + Y_{12}V_2 + Y_{13}V_3\right] \quad \text{..........} (2)$$

$$S_2 = V_2\left[Y_{21}V_1 + Y_{22}V_2 + Y_{33}V_3\right] \quad \text{..........} (3)$$

$$S_3 = V_3\left[Y_{31}V_1 + Y_{32}V_2 + Y_{33}V_3\right] \quad \text{..........} (4)$$

## 1.10.11 Gauss-Seidel Method of Calculating bus voltages of the Network

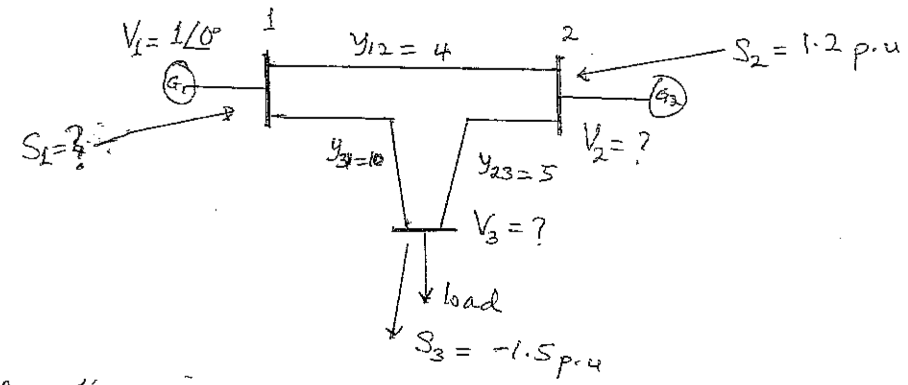
The bus voltages are as given by the following equations;

$$V_1 = \frac{1}{Y_{11}}\left[\frac{S_1^*}{V_1^*} + Y_{12}V_2 + Y_{13}V_3\right] \quad \text{..........} (5)$$

$$V_2 = \frac{1}{Y_{22}}\left[\frac{S_2^*}{V_2^*} + Y_{21}V_1 + Y_{23}V_3\right] \quad \text{------- (6)}$$

$$V_3 = \frac{1}{Y_{33}}\left[\frac{S_3^*}{V_3^*} + Y_{31}V_1 + Y_{32}V_2\right] \quad \text{------- (7)}$$

## Exercise




Given the interconnected power system shown above with;

Bus 1 is slack bus: $V_1 = 1\angle 0°$

Scheduled power at Bus 2: 1.2 p.u.

Scheduled power at Bus 3: -1.5 p.u.

Compute:

(1) $Y_{bus}$ model,

(2) Bus voltages using Gauss-Siedel method,

(3) Power supplied by the swing bus.

## Solution:

(1) $$Y_{bus} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & Y_{23} \\ Y_{31} & Y_{32} & Y_{33} \end{bmatrix} = \begin{bmatrix} 14 & -4 & -10 \\ -4 & 9 & -5 \\ -10 & -5 & 15 \end{bmatrix}$$

To start the Gauss-Siedel iterations, since bus 1 voltage is given as $V_1 = 1$, then, assume that the unknown voltages $V_2 = V_3 = V_1 = 1$. The iterations table is as shown in Table I.

Table I: Gauss-Siedel Iterations Results

| Iterations | $V_2$ | $V_3$ |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 1.1333 | 0.9 |
| 2 | 1.0621 | 0.9333 |
| 3 | 1.0885 | 0.9136 |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |

$$V_2^1 = \frac{1}{9}\left[\frac{-1.2}{-1} + (4\times1) + (5\times1)\right] = \frac{1}{9}\left[1.2+4+5\right] = \frac{1}{9}\left[10.2\right] = \underline{\underline{1.1333}}$$

$$V_3^1 = \frac{1}{15}\left[\frac{1.5}{-1} + (10\times1) + (5\times1)\right] = \frac{1}{15}\left[-1.5+10+5\right] = \frac{1}{15}\left[13.5\right] = \underline{\underline{0.9}}$$

$$V_2^2 = \frac{1}{9}\left[\frac{-1.2}{-1.1333} + (4\times1) + (5\times0.9)\right] = \frac{1}{9}\left[1.0588+4+4.5\right] = \frac{1}{9}\left[9.5588\right] = \underline{\underline{1.0621}}$$

$$V_3^2 = \frac{1}{15}\left[\frac{1.5}{-0.9} + (10\times1) + (5\times1.1333)\right] = \frac{1}{15}\left[-1.6667+10+5.6665\right] = \frac{1}{15}\left[13.9998\right]$$

$$= \underline{\underline{0.9333}}$$

$$V_2^3 = \frac{1}{9}\left[\frac{-1.2}{-1.0621} + (4\times1) + (5\times0.9333)\right] = \frac{1}{9}\left[1.1298+4+4.6665\right]$$

$$= \frac{1}{9}\left[9.7963\right] = \underline{\underline{1.0885}}$$

$$V_3^3 = \frac{1}{15}\left[\frac{1.5}{-0.9333} + (10\times1) + (5\times1.0621)\right] = \frac{1}{15}\left[-1.6072+10+5.3105\right]$$

$$= \frac{1}{15}\left[13.7033\right]$$

$$= 0.9136.$$

The iterations converges at 1.078 for $V_2$ and 0.913 for $V_3$.

Hence, $V_2 = \underline{\underline{1.078\ p.u}}$, $V_3 = \underline{\underline{0.913\ p.u}}$.

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3 Power supplied by the swing bus is;

$$S_1 = V_1\left[y_{11}V_1 + y_{12}V_2 + y_{13}V_3\right]$$

$$= 1\times\left[(14\times 1) - (4\times 1.078) - (10\times 0.917)\right]$$

$$= \left[14 - 4.312 - 9.17\right] = \underline{\underline{0.518\ p.u}}$$

# 2.0 Faults in Interconnected Power Systems

## 2.1 Generators

Some of the important faults which may occur on an alternator are:

1. failure of prime-mover
2. failure of field
3. overcurrent
4. overspeed
5. overvoltage
6. Unbalanced loading
7. Stator winding faults

**2.1.1** Failure of prime-mover: when input to the prime-mover fails, the alternator runs as a synchronous motor and draws some current from the supply system. This motoring condition is known as "inverted running". For gas ~~power~~ and hydro-sets protection against inverted running ~~is achieved~~ can be achieved by ~~dis~~connecting the generator system from the system without harm. But diesel engine system, protective device should be incorporated to solve this problem.

**2.1.2** Failure of field: The chances of field failure ~~of~~ alternators are undoubtedly very rare. Even if it does occur, no immediate damage will be caused, the remedy is for the control room attendant to disconnect the the faulty alternator manually from the system bus-bar.

**2.1.3** Overcurrent: It occurs mainly due to partial breakdown of winding insulation or due to overload on the supply system. Overcurrent protection is not considered ~~too~~ necessary for alternators. The modern tendency is to design ~~or to design~~ alternators with very high values of internal impedance so that they will stand a complete short-circuit at their terminals for sufficient time without serious overloading.

2.1.4 <u>Overspeed</u>: The chief cause of overspeed is the sudden loss of all or the major part of load on the alternator. Modern alternators are usually provided with mechanical centrifugal devices mounted on their driving shafts to trip the main valve of the prime-mover when a dangerous overspeed occurs.

2.1.5 <u>Over-voltage</u>: The field excitation system of modern alternators is so designed that over-voltage conditions at normal running speeds cannot occur. However, over-voltage in an alternator occurs when speed of the prime-mover increases due to sudden loss of the alternator load.

2.1.6 <u>Unbalanced Loading</u>: Unbalanced loading means that there are different phase currents in the alternator. Unbalanced load arises from faults to earth or faults between phases on the circuit external to the alternator. The unbalanced currents, if allowed to persist, may either severely burn the mechanical fixings of the rotor core or damage the field winding. A protective device is encorporated in the generator to trip the circuit breaker to disconnect the alternator from the system.

2.1.7 <u>Stator Winding faults</u>: These faults occur mainly due to the insulation failure of the stator windings. The main types of stator winding faults, in order of importance are;

## 2.2 Transformers

Transformers are static devices, totally enclosed and generally immersed. Therfore, chances of faults occuring on them are very rare. However, the consequences of even a rare fault may be very serious unless the transformer is quickly disconnected from the system. This necessitates to provide adequate automatic protection for transformers against possible faults. Common transformer faults are;

1. Open circuits,
2. Overheating, and
3. Winding short-circuits, eg. earth-faults, phase-to-phase faults and inter-turn faults.

(a) Open circuit: An open circuit in one phase of a 3-phase transformer may cause undesirable heating. In practice, relay protection is not provided against open circuits because the condition is relatively harmless. On the occurrence of such a fault, the transformer can be disconnected manually from the system by the attendant.

(b) Overheating: This is caused by sustained overloads or short-circuits and very occasionally by the failure of the cooling system. The relay protection is also not provided against this contigency and thermal accessories are generally used to sound an alarm or control the banks of fans.

(c) Winding short-circuits (also called internal faults): This arise from deterioration of winding insulation due to overheating or mechanical injury. When an internal fault occurs, the transformer must be disconnected quickly from the system because a prolonged arc in the transformer may cause oil fire. Therefore, relay protection is absolutely necessary for internal faults.

## 2.3 Transmission line Faults:

Faults occur on transmission lines when two or more conductors that normally operate with a potential difference



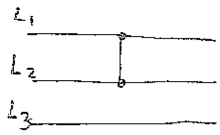
13.3337

come in contact with each other or to earth. These faults may be caused by sudden failure of a piece of equipment, accidental damage or by insulation failure resulting from deterioration or from ~~lighting~~ lightening surges. Irrespective of the causes, the faults on transmission lines can be classified into three categories viz,

1. short circuit fault,
2. open circuit ~~and~~ fault, and
3. earth faults.

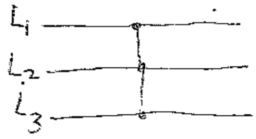
(a) <u>Short Circuit Fault:</u>

The common types of short circuit faults are line-to-line (L-L), three phase short circuit (L-L-L), line-to-line-to-earth (L-L-E), and line-to-line-to-line-to-earth (L-L-L-E), respectively. The various short circuits are shown in Fig. . These faults are as a result of bridge across the lines.



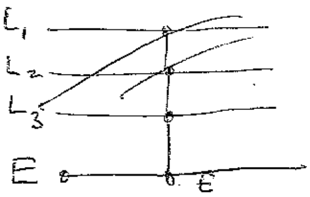
L-L

(a)



L-L-L

(b)

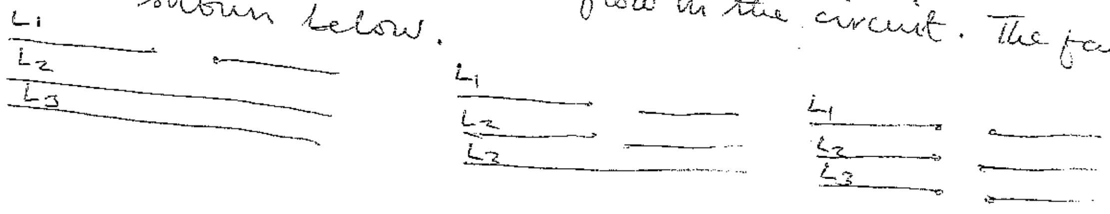


L-L-L-E.

(c)

(b) <u>Open Circuit Fault</u>

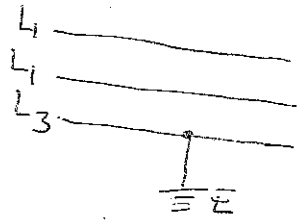
Open conductor conditions may be caused by a blown fuse or by deliberate ~~single~~ selective pole switching operation, or due to physical break in one or two of the three transmission line conductor due to an accident such as a tree fall or a storm. The condition results in an unbalance and thus, asymmetrical currents flow in the circuit. The faults are shown below.



(c) Earth Faults

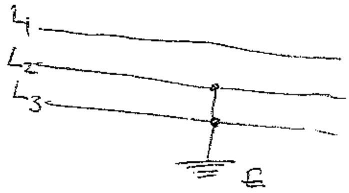
~~When earth fau~~

Earth fault is a condition where ~~the~~ a line or lines bridge to the ground. Under such condition, a heavy current (called short circuit current) flows through the equipment, causing considerable damage to the equipment and interruption of service to the consumers. There is probably no other subject of greater importance to an electrical engineer than the question of determination of short circuit currents under fault conditions. The choice of apparatus and the design and arrangement of practically every equipment in the power system depends upon short-circuit current considerations. Different types of earth faults are shown in the diagrams below.



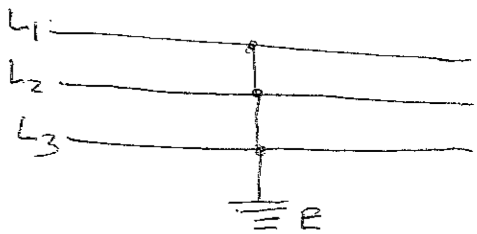

L-E

(a)




L-L-E

(b)

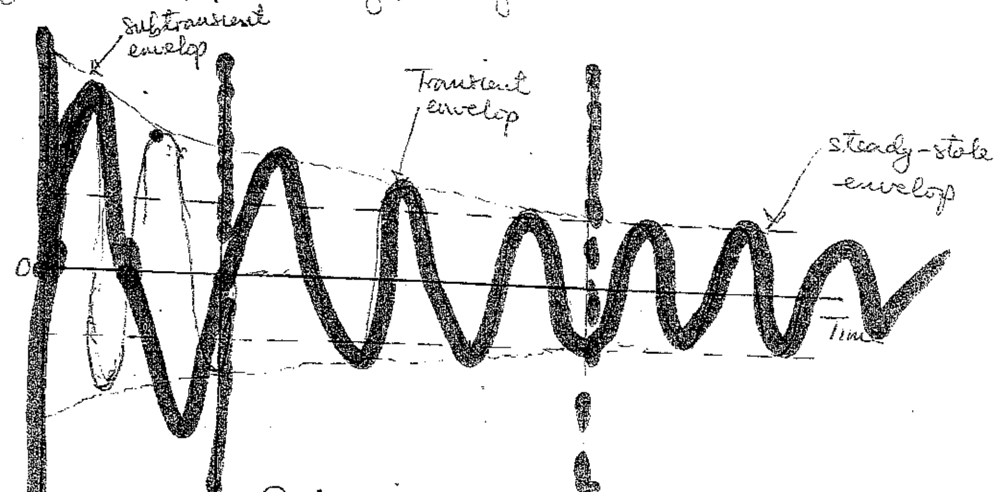



L-L-L-E.

## 2.4 Reactances of a Faulted Generator

The generator exhibits two values of reactances under fault condition, namely, subtransient and transient reactance, respectively. Subtransient reactance is the reactance of the generator during the first cycle of the short-circuit, whereas transient reactance is the reactance of the generator during the the remainder of the short-circuit

The waveform showing subtransient, ~~and~~ transient, and steady state periods of the fault current is shown below.




Symmetrical a.c component of fault current

There are three periods of time:

(a) Subtransient period: first cycle or so after the fault. The current is very large and falls rapidly;

(b) The transient period: current falls at a slower rate;

(c) Steady-state period: current reaches its steady value.

## 2.5 Fault Current Transients in Generators

The a.c current flowing in the generator during the subtransient period is called the subtransient current and is denoted by $I''$. The time constant of the subtransient current is denoted by $T''$ and it can be determined from the envelop. This current can be as much as 10 times the steady-state fault current.

The a.c current flowing in the generator during the transient period is called the transient current and is denoted by $I'$. The time constant of the transient current is denoted by $T'$. This current is often as much as 5 times the steady-state fault current.

After the transient period, the fault current reaches to a steady-state condition $I_{SS}$. This current is obtained by dividing the induced voltage ($E_A$) by the synchronous reactance ($X_s$)

$$I_{SS} = \frac{E_A}{X_s}$$

The rms value of the a.c. fault current in a synchronous generator varies over time as follows,

$$I(t) = \left(I'' - I'\right)e^{-t/T''} + \left(I' - I_{ss}\right)e^{-t/T'} + I_{ss}$$

The subtransient and transient reactances are defined as the ratio of the internally generated voltage to the subtransient and transient current components:

$$X'' = \frac{E_A}{I''} \quad \text{and} \quad X' = \frac{E_A}{I'}$$

<u>Example</u>

A 100 MVA, 13.8 KV, Y-connected, 3-phase 50 Hz synchronous generator is operating at the rated voltage and no load when a 3-phase fault occurs at its terminals. Its reactances per unit to the machine's own base are; $X_s = 1.00$, $X' = 0.25$ and $X'' = 0.12$. ~~and the time constants are $T' = 1.10s$ and $T'' = 0.04s$.~~
Calculate the subtransient, transient, and steady-state currents for the fault

<u>Soln</u>:

The base current of the generator is

$$I_{L\,base} = \frac{S_{base}}{\sqrt{3}\,V_{L\,base}} = \frac{100\times10^6}{\sqrt{3}\times13{,}800} = 4{,}18\cancel{4}\ A$$

The Subtransient current $I'' = \frac{E_A}{X''} = \frac{1.0}{0.12} = 8.333pu$
$= 34{,}865.27$
$= 34{,}900A$

The transient current, $I' = \frac{E_A}{X'} = \frac{1.0}{0.25} = 4pu = 16{,}700A$
$16{,}736$

The steady state current, $I_{ss} = \frac{E_A}{X_s} = \frac{1.0}{1.0} = 1pu = 4.184A$



23902.30

# Typical Waveforms of Short Circuit Currents

The fault currents transients in synchronous generators are illustrated in Fig. 2.6. The resulting current flow in the phases of generators passes through three stages as shown in phase a-to-c in the figure. In phase-a stage the fault current increases from a value below zero and rises above zero and remains steady in magnitude with the d.c. component superimposed on it.

In phase-b stage, the waveform shows that the fault transient current ~~is to~~ has ~~large~~ positive peak and small negative value from the beginning of the fault which reduces in level with time after which it remains steady with the d.c component superimposed on it.

In phase-c which is the final stage of the transient fault current, it has ~~both~~ equal large positive and negative ~~peaks~~ opposite peaks which reduce with time and finally remains steady with the d.c component superimposed on it.

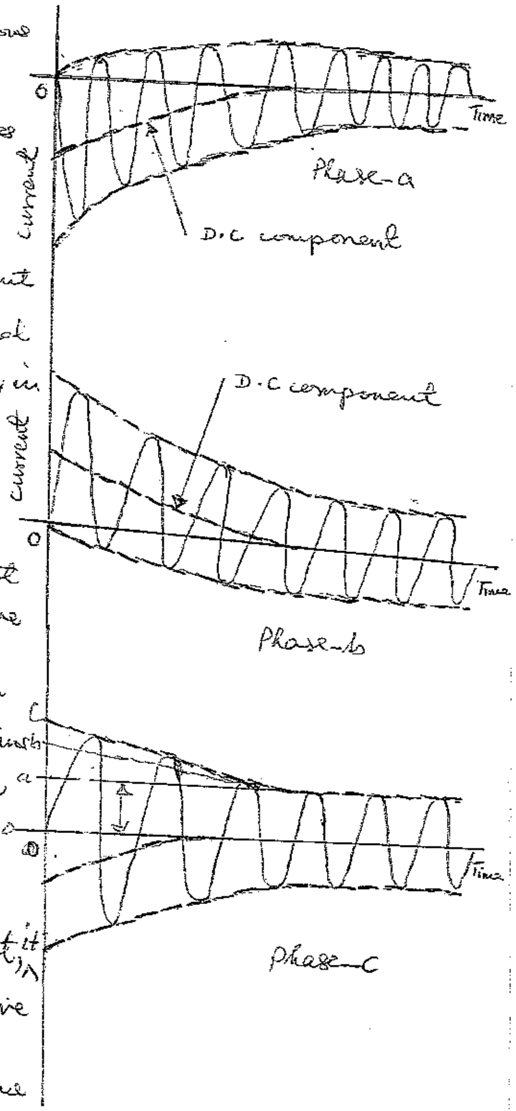



Fig. 2.6 Waveforms of 3-phase fault current transients in synchronous generators.

From phase-c stage of the short circuit fault current waveform, the following definitions are obtained;

1. oa – peak value of steady state short circuit current
2. ob – peak value of transient short circuit current
3. oc – peak value of subtransient short circuit current.

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## Effects of a Travelling Wave on a Transmission System

The effects of a travelling wave on a transmission system are;

1. Travelling wave surges will shatter the insulators and may even wreck poles.
2. When the travelling wave surges hit the windings of a transformer or generator, it may cause considerable damage. The insulation of windings opposes any sudden passage of electric charge through it. Therefore, the electric charges "pile up" against the transformer (or generator). This induces ~~a~~ such excessive pressure between the windings that will cause insulation breakdown, resulting in the production of arc. The initiated arc is sustained by the line voltage long enough to severely damage the machine.
3. If the arc due to voltages is initiated in any part of the power system, this will set up very disturbing oscillations in the line and will generate high voltages that may damage other equipment connected to the line.

## Surge Velocity V, in a uniform line

The surge velocity V, in a uniform line can be obtained from the expression;

$$LC = \frac{1}{V^2}$$

$$\text{or} \quad V = \frac{1}{\sqrt{LC}} \ \text{m/s} \quad \text{------------} \quad (3.8)$$

where L = inductance of the line per unit length.
C = capacitance of the line per unit length.
V = surge velocity

But $L = \frac{\mu_0}{2\pi}\log_e\left(\frac{d}{r}\right)$ H/M and $C = \frac{2\pi\varepsilon_0\varepsilon_r}{\log_e\left(\frac{d}{r}\right)}$

where $\mu_0$ = permeability of free space $= 4\pi \times 10^{-7}$

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$\varepsilon_0$ = permittivity of free space = $8.854 \times 10^{-12}$
$\varepsilon_r$ = permittivity of dielectric material.
$d$ = spacing of conductors = 3.6m for overhead lines
$r$ = radius of conductors = 2.5cm.

Substituting for L and C into eqtn (3.8), the expression for surge velocity can be obtained as:

$$V = \frac{1}{\sqrt{\frac{\mu_0}{2\pi}\log_e\left(\frac{d}{r}\right) \cdot \frac{2\pi\varepsilon_0\varepsilon_r}{\log_e\left(\frac{d}{r}\right)}}}$$

$$= \frac{1}{\sqrt{\mu_0\varepsilon_0\varepsilon_r}}$$

Since the medium is air $\varepsilon_r = 1$ for overhead lines, then,

$$\text{Surge Velocity } V = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \text{ m/s} \quad \text{------} (3.9)$$

Substituting for the values of $\mu_0$ and $\varepsilon_0$ into equation (3.9) gives surge velocity as;

$$V = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12}}}$$

$$= \frac{1}{\sqrt{1.112622645 \times 10^{-17}}}$$

$$= \frac{1}{3.333560557 \times 10^{-9}}$$

$$= 2.9997 \times 10^{8} \text{ m/s}$$

$$\simeq 3 \times 10^{8} \text{ m/s (velocity of Light)}.$$

The surge velocity in a single-phase concentric cable is

$$V = \frac{1}{\sqrt{\mu_0\varepsilon_0\varepsilon_r}} = \frac{\text{velocity of Light}}{\sqrt{\varepsilon_r}} \quad \text{---} (3.10)$$

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where $\varepsilon_r$ = permittivity of cable dielectric material.
For commercial cables, $\varepsilon_r$ lies between about 2.5 and 4.0.

**Surge Impedance (or Characteristic Impedance) $Z_0$**

Consider a transmission line having a resistance per unit length R, and inductance per unit length L, a conductance per unit length G, and a capacitance per unit length C. The series impedance of the line $Z = R + j\omega L$ and shunt admittance $Y = G + j\omega C$, per unit length. Then, the surge impedance of the line $Z_0$ is

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

If R and G are negligible, the characteristic impedance of the line $Z_0$ becomes,

$$Z_0 = \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}} \quad \text{------} \quad (3.11)$$

Substituting for L and C into eqtn (3.11) gives,

$$Z_0 = \sqrt{\frac{\frac{\mu_0}{2\pi}\log_e\left(\frac{d}{r}\right)}{\frac{2\pi\varepsilon_0}{\log_e\left(\frac{d}{r}\right)}}}$$

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}\left(\frac{\log_e\left(\frac{d}{r}\right)}{2\pi}\right)^2} \quad \text{--------} \quad (3.12)$$

Substituting for $\mu_0$, $\varepsilon_0$, $d = 3.6m$ and $r = 2.5cm = 0.025m$ into eqtn (3.12), we obtain;

$$Z_0 = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}\left(\frac{\log_e\left(\frac{3.6}{0.025}\right)}{2\pi}\right)^2}$$

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$$Z_0 = \sqrt{141928.739\left(\frac{\log_e(144)}{2\pi}\right)^2}$$

$$= \sqrt{141928.739\left(\frac{4.969813}{2\pi}\right)^2}$$

$$= \sqrt{141928.739(0.79097)^2}$$

$$= \sqrt{88795.379}$$

$$= 297.98\,\Omega$$

$$\simeq 300\,\Omega.$$

Overhead lines surge impedance (characteristic impedance) lies between 300Ω and 600Ω. While for armoured cables the characteristic impedance $Z_0$ lies between 50Ω and 90Ω.

<u>Reflected and Transmitted Surge Voltages and Currents on Transmission Lines</u>

Due to a variety of reasons, such as a direct stroke of lightning on the line, or by switching operations or by faults, high voltage surges are induced on the transmission line. The surge travels along the overhead line at approximately the speed of light. These waves, as they reach end of the line or junction of transmission lines, part of the incident wave is reflected back, and a part of it is transmitted beyond the junction or termination.

The incident wave $E_i$, the reflected wave $E_r$, and the transmitted wave $E_t$ are as shown on the diagram below.

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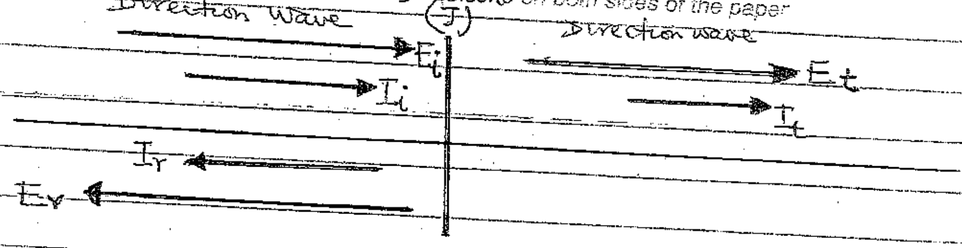
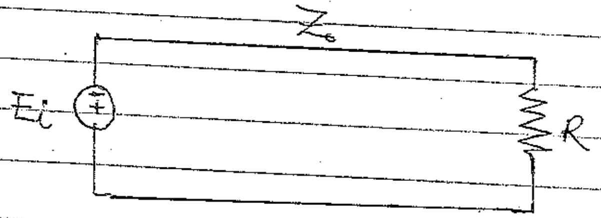



(Reflection at a Junction)

Expression for Reflected and Transmitted Surge Voltages and Currents




Consider a lossless transmission line which has a surge impedance of $Z_o$ terminated through a resistance $R$ as shown.
Let;

$E_i$ = incident voltage wave,
$I_i$ = incident current wave,
$E_r$ = reflected voltage wave,
$I_r$ = reflected current wave,
$E_t$ = transmitted voltage wave,
$I_t$ = transmitted current wave;

The incident wave, the reflected wave, and the transmitted wave obey Kirchhoff's voltage law.
Hence,

$$I_r = -\frac{E_r}{Z_0} \quad \text{--------} \quad (3.13)$$

$$I_i = \frac{E_i}{Z_0} \text{ or } E_i = Z_0 I_i \quad \text{--------} \quad (3.14)$$

$$I_t = \frac{E_t}{R} \quad \text{--------} \quad (3.15)$$

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From Kirchhoff's current Law,

$$I_t = I_i + I_r \quad \text{-------------} \quad (3.16)$$

Also from Kirchhoff's voltage,

$$E_t = E_i + E_r \quad \text{-------------} \quad (3.17)$$

Substituting $I_t$, $I_i$ and $I_r$ into eqtn (3.16) gives,

$$\therefore \frac{E_t}{R} = \frac{E_i}{Z_0} - \frac{E_r}{Z_0} \quad \text{-------------} \quad (3.18)$$

or $\frac{E_t}{R} = \frac{E_i}{Z_0} - \frac{E_t - E_i}{Z_0}$ (from eqtn (3.17) $E_r = E_t - E_i$)

$$\therefore \frac{E_t}{R} = \frac{E_i - (E_t - E_i)}{Z_0} = \frac{E_i - E_t + E_i}{Z_0} = \frac{2E_i - E_t}{Z_0}$$

That is

$$\frac{E_t}{R} = \frac{2E_i - E_t}{Z_0} = \frac{2E_i}{Z_0} - \frac{E_t}{Z_0}$$

or $\frac{E_t}{R} + \frac{E_t}{Z_0} = \frac{2E_i}{Z_0}$

$$E_t\left(\frac{1}{R} + \frac{1}{Z_0}\right) = \frac{2E_i}{Z_0}$$

$$\therefore E_t\left(\frac{Z_0 + R}{RZ_0}\right) = \frac{2E_i}{Z_0}$$

Hence,

$$E_t = \cancel{\frac{\frac{2E_i}{Z_0}}{\frac{Z_0+R}{RZ_0}}} = \frac{2E_i}{\cancel{Z_0}} \times \frac{R\cancel{Z_0}}{Z_0 + R}$$

$$\therefore E_t = \frac{2RE_i}{Z_0 + R}$$

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$\therefore$ Transmitted voltage wave $E_t = \dfrac{2R}{Z_0+R} \cdot E_i$ ------ (3.19)

And transmitted current $I_t$ is from eqtn (3.15) gives,

$$I_t = \frac{E_t}{R} = \cfrac{\cfrac{2RE_i}{Z_0+R}}{R}$$

$$\therefore I_t = \frac{2RE_i}{Z_0+R} \times \frac{1}{R} = \frac{2E_i}{Z_0+R} \quad \text{------ (3.20)}$$

From eqtn (3.14) $E_i = Z_0 I_i$ and substituting for $E_i$ into eqtn (3.20) gives transmitted current wave as

$$I_t = \frac{2Z_0}{Z_0+R} \cdot I_i \quad \text{------ (3.21)}$$

Hence, transmission coefficient for current wave is

$$t_c = \frac{2Z_0}{Z_0+R} \cdot \cancel{I_i} \quad \text{------ (3.22)}$$

Also transmission coefficient for voltage wave is

$$t_v = \frac{2R}{Z_0+R} \cdot \cancel{E_i} \quad \text{------ (3.23)}$$

Similarly, from eqtn (3.18) which is

$$\frac{E_t}{R} = \frac{E_i}{Z_0} - \frac{E_r}{Z_0}$$

From eqtn (3.17), $E_t = E_i + E_r$ and substituting for $E_t$ into the above equation gives,

$$\frac{E_i + E_r}{R} = \frac{E_i}{Z_0} - \frac{E_r}{Z_0}$$

or $$\frac{E_i}{R} + \frac{E_r}{R} = \frac{E_i}{Z_0} - \frac{E_r}{Z_0}$$

Collecting like terms together as follows,

$$\frac{E_r}{R} + \frac{E_r}{Z_0} = \frac{E_i}{Z_0} - \frac{E_i}{R}$$

$$E_r\left(\frac{1}{R} + \frac{1}{Z_0}\right) = E_i\left(\frac{1}{Z_0} - \frac{1}{R}\right)$$

$$E_r\left(\frac{Z_0 + R}{R Z_0}\right) = E_i\left(\frac{R - Z_0}{Z_0 R}\right)$$

Hence,

$$E_r = E_i\left(\frac{R - Z_0}{\cancel{Z_0 R}} \times \frac{\cancel{R Z_0}}{Z_0 + R}\right) = \frac{R - Z_0}{(Z_0 + R)} . E_i$$

The reflected voltage wave ~~a~~ $E_r$ is given as

$$E_r = \frac{R - Z_0}{Z_0 + R} . E_i \text{ ---------- } (3.24)$$

From eqtn (3.13) $I_r = -\frac{E_r}{Z_0}$, substituting for $E_r$ into this equation gives,

$$I_r = -\left(\frac{R - Z_0}{Z_0 + R}\right)\frac{E_i}{Z_0} \text{ ---------- } (3.25)$$

From eqtn (3.14), $I_i = \frac{E_i}{Z_0}$, then, substituting for $I_i$ into eqtn (3.25) gives, the reflected current wave as.

$$I_r = -\left(\frac{R - Z_0}{Z_0 + R}\right) I_i \text{ ---------- } (3.26)$$

Hence, the reflection coefficient $\rho$ for current waves is

$$\rho_{c} = -\left(\frac{R - Z_0}{Z_0 + R}\right) \text{ ---------- } (3.27)$$

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and the reflection Coefficient $\rho$ for voltage wave is

$$\rho_v = +\left(\frac{R - Z_0}{R + Z_0}\right) \quad \text{--- (3.27)}$$

Example 1:

A cable has an inductance $L = 2.41 \times 10^{-7}$ H/M and capacitance $C = 0.1846 \times 10^{-9}$ F/M. Find

(a) velocity of propagation,

(b) surge impedance.

Solution:

(a) velocity of wave propagation is,

$$V = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.41 \times 10^{-7} \times 0.1846 \times 10^{-9}}} \quad \frac{1}{\cancel{4.44886} \times 10^{-17}}$$

$$= \underline{1.5 \times 10^{8} \text{ m/s}}$$

(b) Surge Impedance of the cable is,

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{2.41 \times 10^{-7}}{0.1846 \times 10^{-9}}}$$

$$= \underline{36.13 \Omega}$$

Example 2:

A wave travels along a 500 Km line terminated with a resistance of 1000Ω. The line has a resistance of 0.3Ω/Km and a surge impedance of 400Ω. If the voltage at the termination point after two successive reflections is 200kV, determine the amplitude of the incoming surge.

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Solution :

Length of the line = 500km.

Termination resistance $R = 1000\,\Omega$

Line resistance $= 0.3\,\Omega/km$

Surge impedance $Z_0 = 400\,\Omega$

Termination voltage, $E_L = 200kV$

The total line resistance for 500km,

$$= 0.3 \times 500 = 150\,\Omega$$

The amplitude of the incoming surge $E_i = \frac{2RE_L}{R+Z_0+150}$

$$= \frac{2 \times 1000 \times 200 \times 10^3}{1000+400+150}$$

$$= \frac{400 \times 10^6}{1550}$$

$$= 258064.51V$$

$$= \underline{258kV}$$

Example 3

An overhead line with inductance and capacitance per Kilometer length of 1.3mH and 0.09μF, respectively is connected in series with an underground cable having inductance and capacitance of 0.2mH/km and 0.3μF/km, respectively. Calculate the values of reflected and refracted (transmitted) waves of voltage and current at the junction due to a voltage surge of 100kV travelling to the junction from the overhead line.

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**Solution :**

The characteristic impedance of the line $Z_{0L} = \sqrt{\frac{L}{C}} = \sqrt{\frac{1.3\times10^{-3}}{0.09\times10^{-6}}}$

$= 120.18\,\Omega$

The characteristic impedance of the cable $Z_{0C} = \sqrt{\frac{L}{C}} = \sqrt{\frac{0.2\times10^{-3}}{0.3\times10^{-6}}}$

$= 25.82\,\Omega$

∴ The voltage wave of magnitude, $E_i = 100KV$ which is initiated in the overhead line is partly reflected and partly transmitted on the cable at the junction of the line and the cable.

(i) Therefore transmitted wave $E_t = \dfrac{2Z_{0C}.E_i}{Z_{0C}+Z_{0L}}$

$$= \frac{2\times25.82\times100\times10^3}{25.82+120.18}$$

$$= \underline{\underline{35.37KV}}$$

(ii) Reflected Voltage $E_r = \dfrac{(Z_{0C}-Z_{0L}).E_i}{Z_{0C}+Z_{0L}}$

$$= \frac{(25.82-120.18)\times100\times10^3}{25.82+120.18}$$

$$= \underline{\underline{-64.63KV}}$$

(iii) Transmitted current, $I_t = \dfrac{2E_i}{Z_{0C}+Z_{0L}} = \dfrac{2\times100\times10^3}{25.82+120.18}$

$$= \frac{200\times10^3}{146}$$

$$= 1.369\times10^3 A$$

$$= \underline{\underline{1.37KA}}$$

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(iv) Reflected Current, $I_r = \dfrac{E_r}{Z_{oL}}$

$= \dfrac{-64.63 \times 10^3}{120.18}$

$= -0.53777 \times 10^3 A$

$= -537.8A$

Protection of Transmission Lines against Lighting Surges

Lighting surges may cause serious damage to the expensive equipment in the power system (e.g. generators, transformers). It is necessary to provide protection against these surges. The most commonly used devices for protection against lighting surges are;

1. earthing screen,
2. overhead ground wires
3. Lightning arresters or surge diverters.

Earthing Screen

The power stations and sub-stations generally house expensive equipment. These stations can be protected against direct lightning strokes by providing earthing screen. It consists of a network of copper conductors (generally called shield or screen) mounted all over the equipment in the sub-station or power station. On the occurrence of direct lightning stroke on the station, screen provides a low resistance path by which lightning surges are conducted to ground.

Overhead Ground Wires

The most effective method of providing protection to transmission lines against lightning strokes is by the use of overhead ground wires as shown in the figure below. For simplicity, one ground wire and three lines are shown. The ground wires are placed above the line conductors at such positions that

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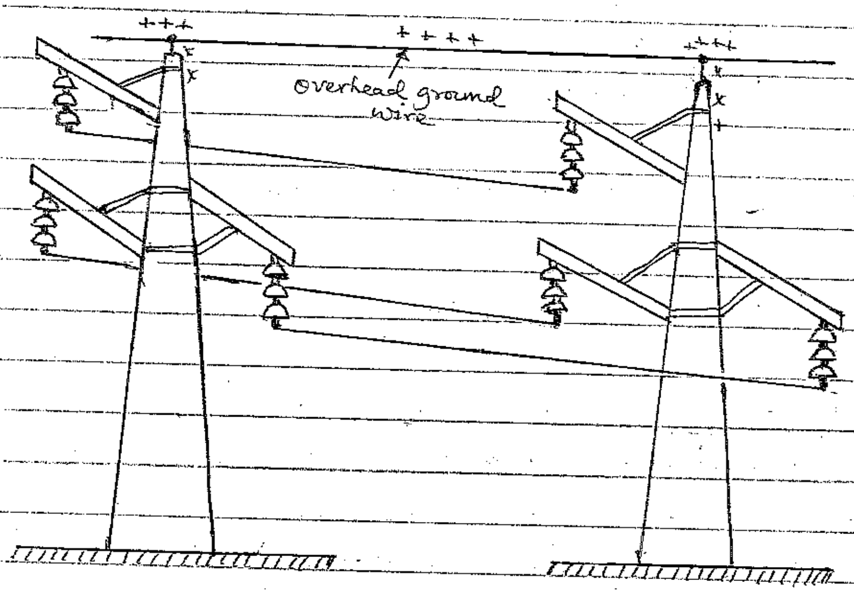
practically all lightning strokes are intercepted by the wires and conducted to the ground through the tower steel body.




Lightning Arresters

The earthing screen and ground wires can well protect the electrical system against direct lightning strokes but they fail to provide protection against travelling waves which may reach the terminal equipment. The lightning arresters or surge diverters ~~that~~ provide protection against such surges. Surge diverters are protective devices which conduct high voltage surges on the power system to the ground.

The figure below shows the basic form of surge diverter. It consists of a spark gap in series with a non-linear resistor. One end of the diverter is connected to terminal of the equipment to be protected and the other end is effectively grounded. The length of the gap is so set that normal line voltage is not enough to cause an arc across the gap but a dangerously high voltage will breakdown the air insulation and form an arc. The property of the non-linear resistance is that its resistance decreases as the

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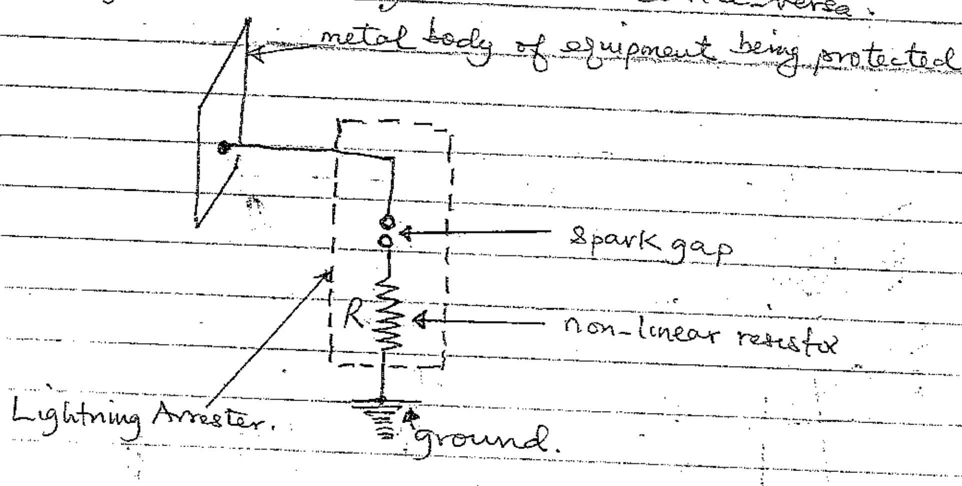
the voltage (or current) increases and vice-versa.




(1) Terminations of Transmission Line at $R = Z_0$

When the end of transmission line is terminated on a junction that maches ~~the~~ characteristic impedance of the line, i.e., $Z_0 = R$ (termination or junction resistance), then there is no reflection and the reflection coefficient $\rho = 0$. When the characteristic impedance of transmission $Z_0 = R$ (termination resistance) the line is said to be matched). Matched load absorbs the incident wave completely, corresponding to transmission coefficient $t = 1$.

(2) Termination of Transmission Line at $R > Z_0$

When a voltage surge $E_i$ arrives at the junction J, which is terminated at resistance R greater than the line characteristic impedance $Z_0$, part of it is reflected back without change in sign. Also, part of the incident current is reflected back with a change in sign. And parts of the incident voltage and current are transmitted across the junction. The diagrams for the voltage and current waves are shown below.

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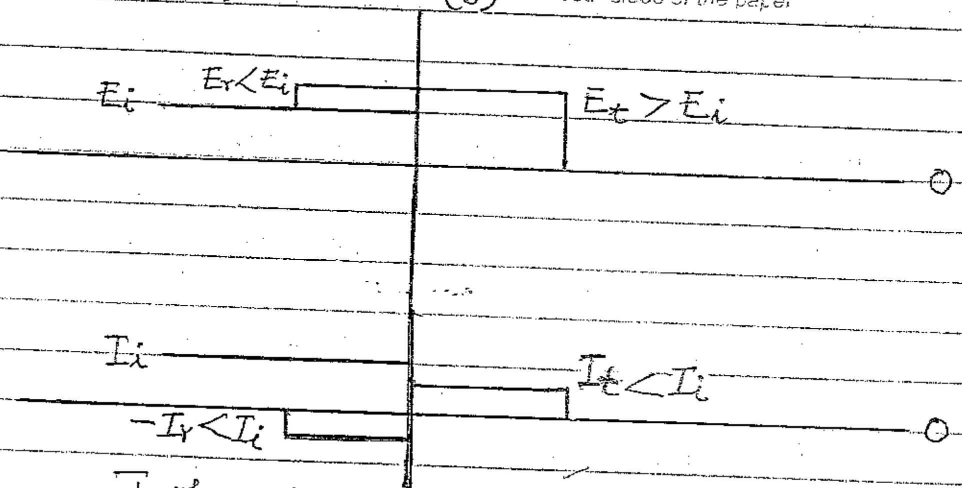
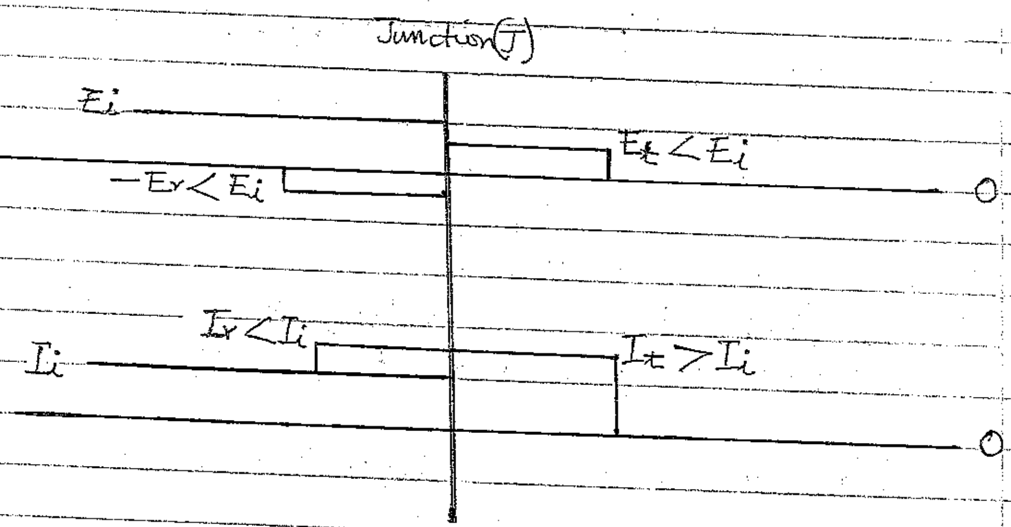



Just after Reflection at Junction.

## (3) Termination of Transmission Line at $R < Z_0$

When a voltage surge $E_i$ arrives at the junction $J$, which is terminated at resistance $R$ less than the line characteristic impedance $Z_0$, part of it is reflected back with change of sign. Also, part of the incident current is reflected back without change of sign. And parts of the incident voltage and current are transmitted across the junction. The diagrams for the voltage and current waves are shown below.




Just after Reflection at Junction

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(4) Termination of Transmission Line at $R = \infty$

When a transmission line with characteristic impedance $Z_0$ is open circuited at the far end (ie. $R = \infty$), then the following expressions hold for reflected voltage and current waves;

For voltage wave of eqtn (3.24), $E_r$ becomes

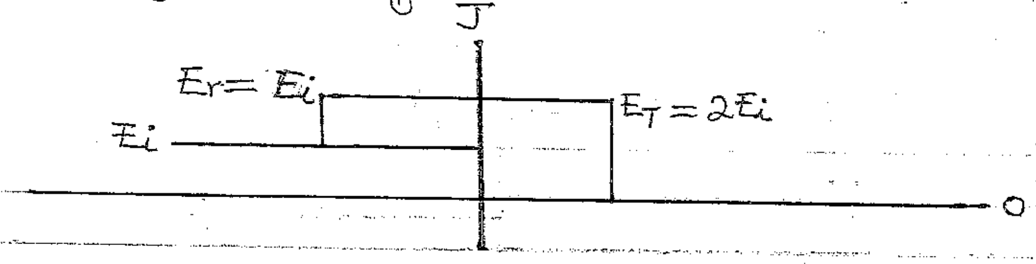
$$E_r = E_i\left(\frac{\infty - Z_0}{Z_0 + \infty}\right) \quad \text{since } R = \infty$$

* correct your note

$$E_r = E_i \cdot 1$$

$$\text{or} \quad E_r \simeq E_i \qquad (3.29)$$

From eqtn (3.29) it can be seen that when a line is terminated at an open circuit, the incident voltage is fully reflected back without change of sign ($E_i$). Hence, the voltage in the line is twice the incident voltage ($2E_i$) because the reflected voltage adds to the incident voltage. This corresponds to reflection coefficient $\rho = 1$. The total voltage at the far end of the line is twice the incident




voltage, corresponding to voltage transmission coefficient $t = 2$. The diagram showing the incident, reflected and transmitted waves at the junction is shown above.

For current wave of eqtn (3.26), $I_r$ becomes;

$$I_r = -\left(\frac{\infty - Z_0}{Z_0 + \infty}\right) I_i \quad \text{since } R = \infty$$

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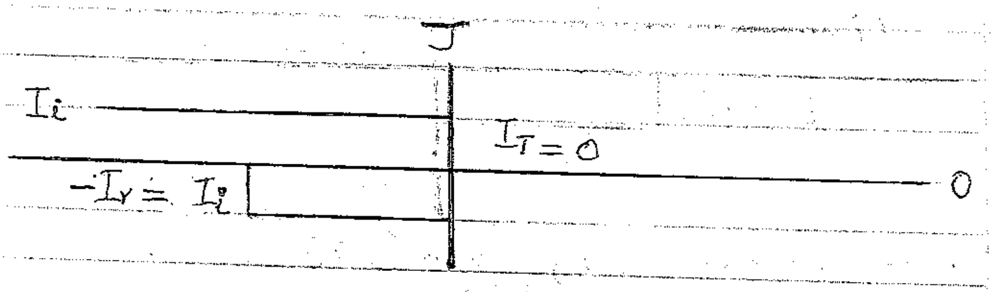
$$I_r = -I_i \cdot 1$$
$$\simeq -I_i \quad \text{------------} \quad (3.30)$$

From eqtn (3.30), it can be seen that when a line is terminated at an open circuit, the incident current is fully reflected back with opposite sign ($-I_i$). Hence, the current in the line is zero because the reflected current subtracts from the incident current. This corresponds to a reflection coefficient $\rho = -1$. The total current at the far end of the line is zero, corresponding to current transmission coefficient $t = 0$. The diagram showing the incident, reflected and transmitted current waves at the junction is shown below.




(5) <u>Termination of Transmission Line at $R = 0$</u>

When a transmission line with characteristic impedance $Z_0$ is short circuited at the far end (i.e., $R = 0$), then the following expressions hold for reflected voltage and current waves:

For voltage wave of eqtn (3.24), $E_r$ becomes

$$E_r = E_i\left(\frac{0 - Z_0}{Z_0 + 0}\right) = E_i\left(\frac{-Z_0}{Z_0}\right) \quad \text{since } R = 0$$

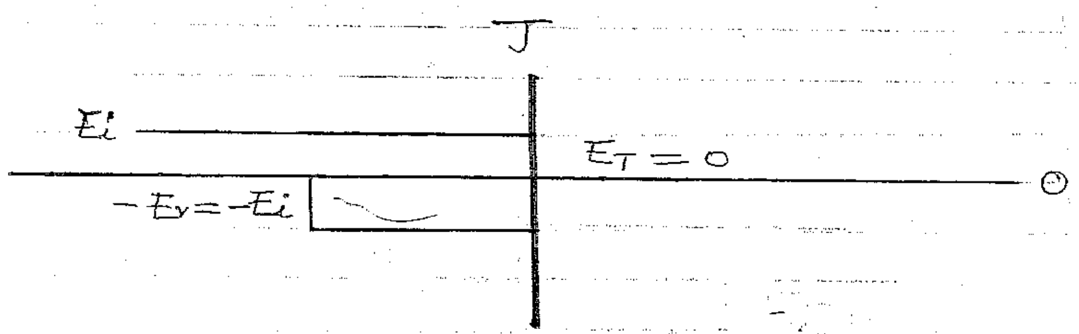
* correct note

$$\therefore E_r = -E_i \cdot 1$$

$$\text{or} \quad E_r = -E_i \quad \text{------------} \quad (3.31)$$

(4)

From eqtn (3.21), it follows that when a line is terminated at a shortcircuit, the incident voltage is fully reflected back with change of sign $(-E_i)$, so as to cancel the incoming surge and the voltage reflection coefficient $\rho = -1$. The total voltage at the end of the line is zero, corresponding to voltage transmission coefficient $t = 0$. Hence, the voltage in the line is zero because the reflected voltage is subtracted from the incident voltage. The diagram showing the incident, reflected and transmitted waves at the junction is shown below.




From current wave of eqtn (3.26), $I_r$ becomes;

$$I_r = -\left(\frac{0 - Z_0}{Z_0 + 0}\right) \cdot I_i \quad \text{since } R = 0$$

$$= -\left(\frac{-Z_0}{Z_0}\right) \cdot I_i$$

$$= -(-1) \cdot I_i$$

$$\therefore \quad I_r = I_i \qquad \text{------} \quad (3.32)$$

From eqtn (3.22), it follows that when a line is terminated at a short circuit, the incident current is fully reflected back with no change of sign $(I_i)$, so to add to the incoming surge current. Hence the

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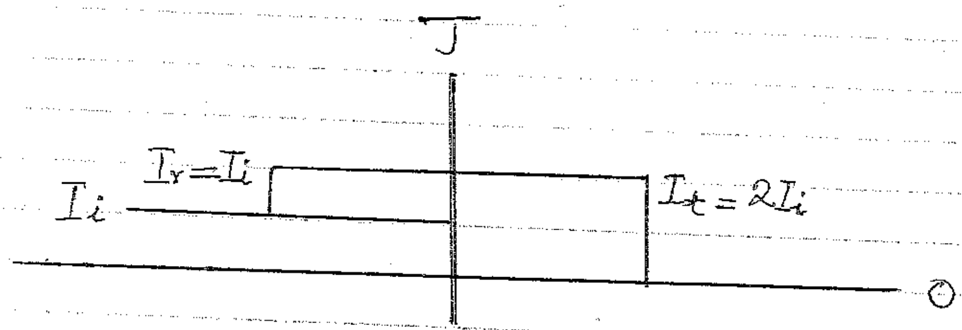
current in the line is twice the incident current ($2I_i$). This corresponds to current reflection coefficient $\rho = 1$. The total current at ~~the~~ far-end of the line is twice the incident current, corresponding to current transmission coefficient $t = 2$. The diagram showing the incident, reflected and transmitted waves at the junction is shown below.

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